

Spectrum

The spectrum is also known as Electromagnetic spectrum. It is defined as.

"The electromagnetic spectrum is the complete spectrum of all forms of light"

- The electromagnetic wave consists of electric and magnetic fields which vibrate.

- The electromagnetic spectrum consist of

Radio wave $\rightarrow$ Cell Phone waves $\rightarrow$ Microwaves $\rightarrow$ radar waves $\rightarrow$ ~~infrared~~ ^{light waves} Infrared waves $\rightarrow$ Visible light waves $\rightarrow$ ultraviolet

Waves : rays $\rightarrow$ X-rays waves $\rightarrow$ gamma waves

Electromagnetic rays form waves.

- The waves have following properties.

1) speed = c

2) frequency = ν

3) wave length = λ

- speed, frequency and wavelength are related to each other in following formula.

$$c = f \times \lambda$$

- All Light waves travel at a speed of $3 \times 10^8 \text{ m/s}$ in a vacuum

Relationship between wavelength and frequency

All parts of electromagnetic spectrum travel at the same ~~the~~ speed

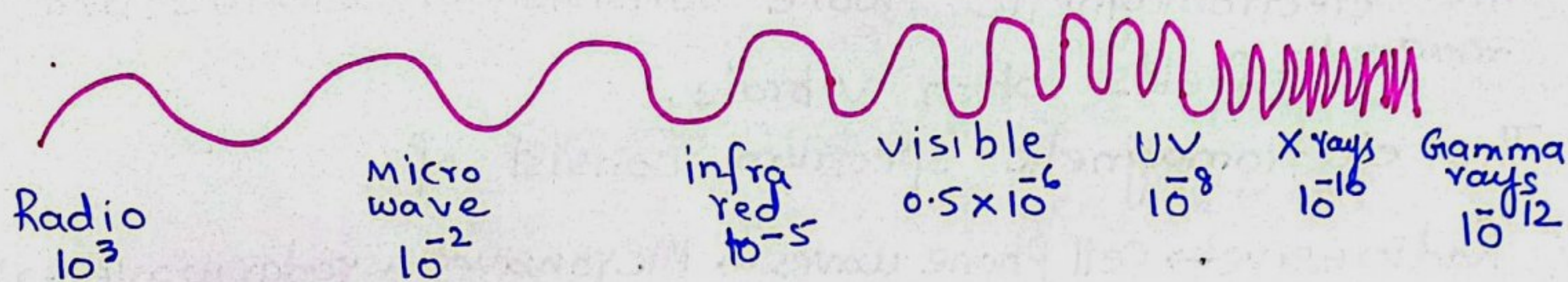
wave length and frequency have an inverse relationship this means as the one characteristic increases, the other decreases. it means that when wavelength increases the frequency decreases

$$\nu = \frac{1}{\lambda}$$

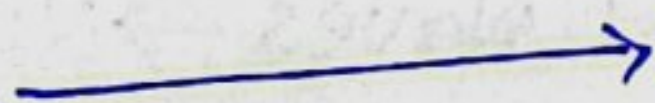
So Short wave length have a high frequency and Long wave lengths have a low frequency

Frequency and energy relationship

Frequency and energy have direct relationship. It means that if electromagnetic waves have high frequency their energy will also be high and waves with low frequency have low energy.



Longer wavelength
Low frequency
Low energy



Shorter wavelength
high frequency
high energy.

characteristics and uses of various Electro magnetic radiations.

1: Micro waves :

- Longer than radio waves, shorter than infrared or visible
- wave length = 1×10^{-4} m to 1×10^{-1} m.
- Used for communication, medicine and microwave oven.

2: Radio waves.

- Low energy waves with long wavelengths.

- It includes FM, AM, radar and TV waves.

- Wave length 10^{-1} m

- Low frequency.

- Used in many devices such as remote control items, cell phones, wireless devices

3: Infra red waves:

- Invisible electromagnetic waves that are detected as heat

- Can be detected with special devices such as night goggles.

- Used in heat lamps

- Higher energy than microwaves but lower than visible light

4: Visible Light

- This portion of electromagnetic spectrum that human eyes can detect
- It includes red, orange, yellow, green, blue, indigo, violet.
- Red has lowest energy and violet has highest energy in visible spectrum.

5: Ultraviolet waves

- Higher energy than visible light.
- Can cause skin cancer and blindness in humans.
- These are using in sterilizing equipment

6: X-rays

- These are high energy waves.
- It can cause cancer.
- Used in medicine, industry and astronomy

7: Gamma rays

- Highest energy rays.
- Blocked from earth's surface by atmosphere

Types of spectrum.

There are two types of spectrum

- (i) Continuous spectrum
- (ii) Line spectrum.

1 Continuous Spectrum.

- In this spectrum the boundary line between the colours cannot be marked.
- The colours diffuses into each other without any dark line
- Best example of continuous spectrum is rain bow.

2: Line spectrum

When an element or its compound is volatilized on a flame and light emitted is seen through a spectrometer we see distinct lines separated by dark line (spaces). This type of spectrum is called line spectrum or atomic spectrum.

There are two ways in which an atomic spectrum can be view.

1: Atomic Emission spectrum

- When Solids are volatilized or element in their gaseous states are heated to high temperature or subjected to an electrical discharge
- Radiation of certain wavelengths are emitted
- This spectrum contained bright lines against a dark background. This is called atomic emission spectrum

2: Atomic Absorption spectrum

- When a beam of white light is passed through a gaseous sample of an element, the element absorbs certain wave lengths while the rest of wavelengths pass through it.
- The spectrum of this radiation is called an atomic absorption spectrum
- The wavelengths of radiation that has been absorbed by the element appear as dark lines and background is bright

Green Chemistry.

- Green chemistry is also known as sustainable chemistry.

Defination: It is a branch of chemistry that deals with the design and optimization of processes and products in order to lower or remove altogether, the production and use of toxic substances.

Green chemistry is not the same as environmental chemistry.

Development of sustainable practices are environment friendly such as a reduction in the consumption of non-renewable resources and strategies to control environmental pollution.

Key Principles of Green Chemistry

The 12 Key principles put forward by the American chemist Paul Anastas and John Warner in the year 1998 to lay the foundation for green chemistry.

These rules are also as follows.

1: Prevention of waste.

Preventing the formation of waste products is always preferable to the clean-up of the waste once it is generated.

2: Atom Economy:

The synthetic processes and methods that are devices through green chemistry must always try to maximise the consumption and incorporation of all the raw materials into final products. This must strictly be followed in order to minimise the waste generated by any process.

3: Avoiding the generation of hazardous chemicals

Reactions and processes that involve the synthesis of certain toxic substances that cause health problems in human must be optimised in order to prevent the generation of such a substances.

4: Design of safe chemicals

During the design of chemical products, care must be taken to make the chemical as non-toxic to human and the

environment as possible.

5: Design of safe auxiliaries and Solvents

The use of auxiliaries in processes must be avoided to the largest possible extent. Even in the circumstances where their use is extremely necessary, they must be optimized to be as non-hazardous as possible.

6: Energy Efficiency:

The amount of energy consumed by the process must be minimized to the maximum possible extent.

7: Incorporation of renewable feed stock.

The use of renewable feed stock and renewable raw materials must be preferred over the use of non-renewable ones.

Feedstock: It is a raw material used for processing or manufacturing another product. Example: 1) Crude oil to produce gasoline
2) Corn is used to produce ethanol.
3) Soybean is used to produce biodiesel.

Renewable Feed stock

A renewable feed stock is a raw material that can replenish itself on a human timescale ideally within months.

Example: 1) Sugars can be derived from renewable feed stocks like maize, Sugarcane
2) Energy crops such as bamboo, wheat grass

8: Reduction in the generation of derivatives

The unnecessary use of derivatives must be minimized such they tend to require the use of additional reagent and chemicals, resulting in the generation of excess waste.

9: Incorporation of catalysts

In order to reduce the energy requirements of chemical reactions in the process, chemical catalysts and catalytic reagents must be used.

10: Designing the chemicals for degradation

When designing a chemical product, care must be taken during the design process to make sure that chemical is not an environmental pollutant. This can be done by making sure that the chemical breaks down into non-toxic substances.

11: Incorporating real time analysis

Process and analytical methodologies must be developed to the point that they can offer real time data for their monitoring.

This enable the involved parties to stop or control the process before toxic / dangerous substances are formed.

12: Incorporation of safe chemistry for the prevention of accidents

While designing chemical processes it is important to make sure that substances that are used in the process are safe to use. This can help prevent certain workplace accidents such as explosions and fires.

This also help to develop a safer environment for the process to take place in.

Impact of Green chemistry.

1: Use of Green solvent

Many chemical synthesis reactions that are carried out on an industrial scale require large amounts of chemical solvents.

These solvents are also used industrially for degreasing and cleaning purpose.

Many traditional solvents that have been used for such purpose in the past are known to be toxic to human beings. Some such solvents are also known to be chlorinated.

The advancement of green chemistry has brought many alternatives to these toxic solvents. The green solvents that are coming up as alternatives are known to be derived from renewable sources and are also known as biodegradable.

Thus green chemistry has great potential to lower the toxicity of certain industrial environments by developing safer alternatives.

Examples: ① Water A cheap and abundant solvent, it is also better for the environment. It is safer than organic solvents.

② Ethyl Acetate: A byproduct of corn, used as a coating for wood, metals etc.

③ Propylene Carbonate: A CO_2 neutral solvent obtained from propylene oxide and carbon dioxide.

④ Limonene: A non-toxic readily available and safe to handle, can be used as an alternative to toxic organic solvent such as toluene.

Green solvents also known as biosolvents are environmentally friendly alternatives to petrochemical solvents. They are derived from the processing of agricultural crops.

Advantages They have lower environmental impact, reduced toxicity and potential biodegradability.

2: Development of specialised synthesis techniques

The development of specialised synthetic techniques can optimise processes in order to make them more environmentally friendly by making them adhere to the

Principles of green chemistry

Example: An enhanced synthetic technique is the development of the olefin metathesis reaction in

field of organic chemistry.

- Other notable developments in green chemistry include
- 1) The employment of supercritical CO_2 as green solvent.
 2. Incorporating aqueous solutions of hydrogen peroxide to drive relatively clean oxidation reactions.

Example Production of Hydrazine

The most popular method for the production of hydrazine (N_2H_4) was the Olin Rasching process, which involved the use of NH_3 and Sodium hypochlorite.

In green chemistry a more environment-friendly alternative to this process was discovered.

Peroxide Process

In peroxide process for the production of hydrazine ammonia is reacted with H_2O_2 . In this alternate method, water is produced as the only side product.

It can also be noted that the peroxide process does not require any extracting solvents.

Significance:

- 1: Major tool in accomplishing pollution prevention
- 2: Leads to reduction in waste
- 3: Reduced use of energy and other perishable resources
- 4: Carrying out chemical activities leading to safer products